

Problem Set 1

Theis Ingerslev Jensen — Yale SOM
MGT 924 Statistical Foundations

General guidelines

- I refer to programming functions using their R name, but you're free to use whichever programming language you prefer
- I put a premium on a professional presentation of your solution, and the way you present your solution—your presentation score—**counts for up to 10 points**
- Do not include code unless explicitly stated
 - I repeat, do not include code unless I explicitly tell you to. Including codes results in a point deduction from your presentation score.
 - Also, don't include the output from functions such as `summary()` directly in your solution. Treat the solution as a professional product, and visualize the results using tables and, especially, figures!
- I put a premium on brief and to-the-point answers. For open-ended questions, a rule of thumb is to aim for at most three lines of text and only exceed this limit if it's crucial to your argument
- Always describe how you arrived at a solution. For example, it's often a good idea to include the formula(s) you've used to answer a question.
- For open-ended questions (e.g., questions that start with “discuss”), there's not necessarily one correct answer, meaning that the validity of the argument is more important than the conclusion
- The points after each questions give you an indication of the weight of the question in the total score. However, the points are indicative only and the final grade is based on an overall assessment of the solution as well as how it is presented
- If you believe that you lack information to answer a question, you should make—and explicitly state—the assumptions that enable you to provide an answer.

1 Visualize before building a prediction rule (20 points)

For this section, we will use the file called `ans.csv` available from Canvas. This file contains x and y values from four different data sets as indicated by the column `dataset`, with 11 observations in each. The goal is to compute several statistics from each data set and use visualization to evaluate the usefulness of these statistics. For concreteness, we can think of each data set as representing a specific stock you want to trade, with y being its return and x some underlying fundamental. Henceforth I'll refer to a specific `dataset` as a *stock*

1.1. For each stock, compute the following sample statistics:

(1.1.1) The mean of x

(1.1.2) The mean of y

(1.1.3) The variance of x

(1.1.4) The variance of y

(1.1.5) The correlation between x and y

(1.1.6) The R^2 from a regression of y on x using only the statistics from (1.1.1)-(1.1.5)

1.2. For each stock, compute the OLS estimates from a regression of y on x . *Hint: Feel free to use `lm()`*

1.3. Based solely on the statistics computed in the two previous questions how would you trade each of the four stocks? *Hint: In other words, if you saw a high or low x what should you do to make money based on the statistics estimated in 1.1 and 1.2?*

1.4. For each stock, create a scatterplot showing the relationship between x and y . Further, add the relevant OLS prediction rule to the plots

1.5. For each stock, describe how the scatterplot can help you improve your trading strategy relative to your answer in 3. *Hint: Only describe the one improvement you view as the most important for each stock*

2 Predicting house prices (40 points)

In this section, we will try to predict the selling price of a house from its tax-assessed value. To do so, we're going to use linear prediction rules of the form

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x,$$

which we covered in class.

2.1. Data Preparation. Please include your code.

- (2.1.1) Download the file `ct_property_data.csv` from Canvas
 - (2.1.2) Create a new column `assessedvalue_true` that reflects the true market value of the house perceived by the tax assessor. *Hint: As discussed in class, in Connecticut, the assessment value for tax purposes (`assessedvalue`) is 70% of the true market value perceived by the tax assessor*
 - (2.1.3) Express `assessedvalue_true` and `saleamount` in \$1,000 dollars. *Hint: So, e.g., \$10,000 becomes \$10*
 - (2.1.4) Filter the data such that it only includes (1) single family houses (2) from New Haven (3) listed in 2008 (4) with a true assessed value below \$250,000. *Hint: The relevant columns are `residentialtype`, `town`, `listyear`, and `assessedvalue_true`, respectively*
- 2.2. Manual prediction rule: Based on your knowledge of the housing data from class and general economic reasoning, propose a manual linear prediction rule to predict the selling price of a house (`saleamount`) from the taxed-assessed value (`assessed_true`)
- 2.3. Compute the residuals from your manual prediction rule, and create a scatterplot showing `assessedvalue_true` on the x-axis, and the residuals on the y-axis. From this plot, briefly consider how your manual prediction rule could be improved (i.e., should the slope and/or intercept be higher or lower)
- 2.4. Compute the covariance between `saleamount` and `assessed_true`, the variance of `assessed_true`, the mean of `assessed_true`, and the mean of `saleamount`. Use these estimates to compute the OLS estimates of $\hat{\beta}_0$ and $\hat{\beta}_1$
- 2.5. Use numerical optimization to find the OLS estimates (please include your code). Are your estimates the same as in the previous question? If not, state which ones you prefer and why. *Hint: By numerical optimization I mean using `optim()` or a similar functions to find the OLS parameter estimates, as we discussed in class.*
- 2.6. Create a scatterplot with x on the x-axis and y on the y-axis. In this plot, include two lines showing the manual and OLS prediction rules, respectively
- 2.7. In class, we estimated $\hat{\beta}_0^{\text{OLS}} = -4.69$ and $\hat{\beta}_1^{\text{OLS}} = 1.37$ based on housing data from 2020, which differs from the OLS estimates based on the data from 2008 and, perhaps,

from your manual prediction rule. If you were to predict house prices in 2023, which prediction rule would you use? Why?

- 2.8. Redo the steps in 1, except for step d (3), i.e., include houses from all listing years rather than just from 2008. Next, compute $\hat{\beta}_0$ and $\hat{\beta}_1$ within each listing year. Finally, create a figure showing how $\hat{\beta}_0$ and $\hat{\beta}_1$ evolves over time. *Hint: It's up to you to decide which type of figure best conveys the time-evolution in the OLS estimates*

3 Predicting portfolio risk (40 points)

For this section, we're going to use the file `jkp_factors.csv`, which contains excess returns from 13 different US long-short equity factors as well as the excess return on the US stock market. These factors are from the paper *Is there a replication crisis in finance* and are available from jkpfactors.com.

Our goal is to predict the risk (standard deviation) of a portfolio that invests 50% in the Value factor, 25% in the Momentum factor, and 25% in the Quality factor (henceforth The Portfolio)

- 3.1. Download the data from Canvas and retain the date column `eom`, as well as the relevant return columns `market`, `value`, `momentum`, and `quality`. Compute the mean of the return columns.
- 3.2. Compute the sample standard deviations of the return columns
- 3.3. Compute the correlation matrix of the return columns
- 3.4. Compute the standard deviation of The Portfolio based on the sample standard deviation and correlations. *Hint: We covered the relevant formula in class*
- 3.5. Separately, regress `value`, `momentum`, and `quality` on `market`. Report the three market beta estimates. *Hint: The market beta is the slope coefficient from the regression*
- 3.6. Compute the sample variance of the residuals for each of the three regressions
- 3.7. Using the market betas, the residual variances, and the variance of the market return, compute the standard deviation of The Portfolio. *Hint: We covered the relevant formula in class*
- 3.8. Why are the two risk predictions from 4. and 7. different?
- 3.9. Which of the two risk predictions do you prefer?